

4 Greenhouse gases are involved in anthropogenic climate change.

(a) (i) Which of the following are greenhouse gases?

(1)

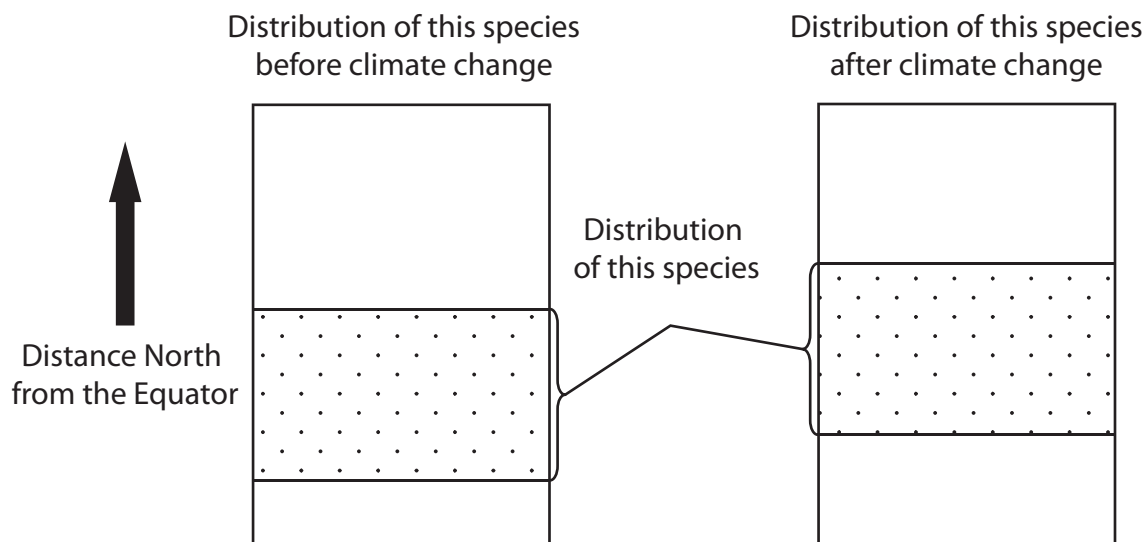
- ☐ **A** carbon dioxide and oxygen
- ☐ **B** methane and water vapour
- ☐ **C** carbon dioxide, oxygen and water vapour
- ☐ **D** methane, carbon dioxide and oxygen

(ii) State what is meant by **anthropogenic climate change**.

(2)

(b) The distribution of species has been affected by climate change.

The diagrams show how the distribution of one species, in a country in the Northern Hemisphere, could change as a result of climate change.



Explain the changes in the distribution of this species in this country.

(3)

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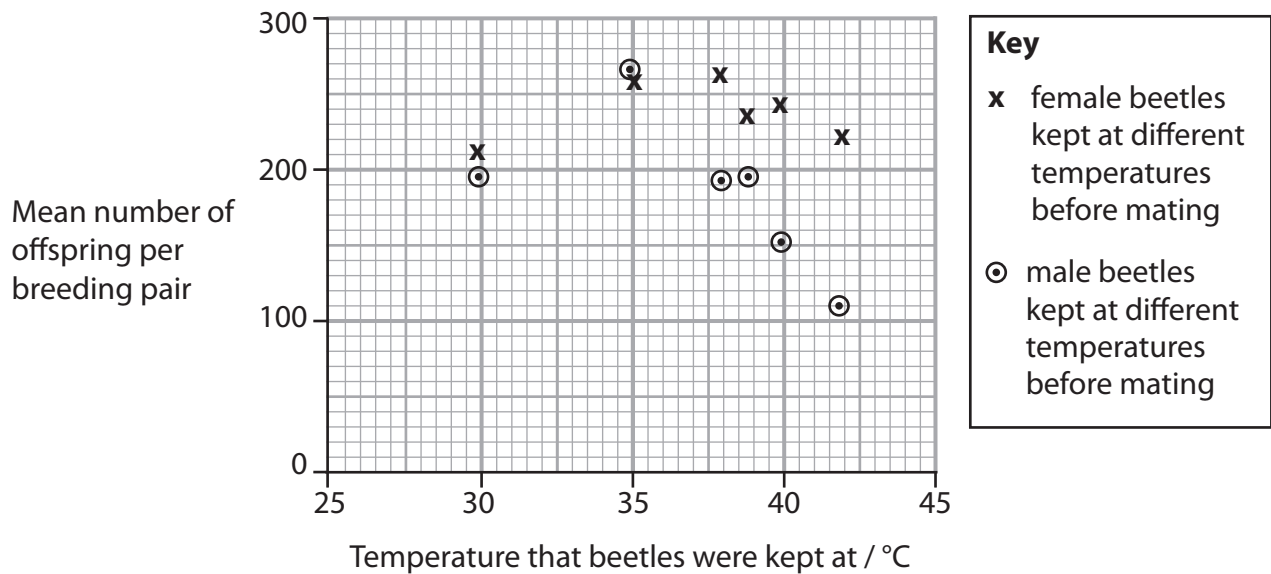
(c) The effect of temperature on reproduction in a species of beetle was studied.

Groups of male and groups of female beetles were each kept at different temperatures for five days.

The beetles were then mated with beetles, of the opposite sex, that had been kept at 30°C for five days.

The mean number of offspring per breeding pair was then determined.

The graph shows the results of this study.



(i) Identify **two** conclusions that can be drawn from this study.

(2)

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(ii) Explain the differences in the results obtained for the male and female beetles kept at different temperatures.

(3)

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(Total for Question 4 = 11 marks)
